



Prayas JEE 2.0 (2024)

Quadratic Equations

DPP-02

- If α and β are the roots of equation $x^2 - 7x + 17 = 0$ and $a_n = \alpha^n + \beta^n$ for $n \geq 1$, then find the value of $\frac{a_{10} + 17a_8}{a_9}$
- Let α & β are the roots of $x^2 - 14x - 2 = 0$ with $\alpha > \beta$. If $S_n = \alpha^n - \beta^n$ for $n \geq 1$, then find the value of $\frac{S_{12} - 2S_{10}}{S_{11}}$
- The least integral value of k for which the equation $x^2 - 2(k+2)x + 12 + k^2 = 0$ has two distinct real roots is
 (1) 0 (2) 2
 (3) 3 (4) 4
- The roots of the given equation $(p-q)x^2 + (q-r)x + (r-p) = 0$ are
 (1) $\frac{p-q}{r-p}, 1$ (2) $\frac{q-r}{p-q}, 1$
 (3) $\frac{r-p}{p-q}, 1$ (4) $1, \frac{q-r}{p-q}$
- If α and β are roots of equation $x^2 - 2x + 3 = 0$, then the equation whose roots are $\frac{\alpha-1}{\alpha+1}$ and $\frac{\beta-1}{\beta+1}$ will be
 (1) $3x^2 - 2x - 1 = 0$ (2) $3x^2 + 2x + 1 = 0$
 (3) $3x^2 - 2x + 1 = 0$ (4) $x^2 - 3x + 1 = 0$
- Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which one of the following statements is not true?
 (1) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$
 (2) $p_5 = 11$
 (3) $p_3 = p_5 - p_4$
 (4) $p_4 p_5 = p_2 \cdot p_3$
- If α, β are the roots of $ax^2 + bx + c = 0$, then the equation whose roots are $2 + \alpha, 2 + \beta$ is
 (1) $ax^2 + x(4a - b) + 4a - 2b + c = 0$
 (2) $ax^2 + x(4a - b) + 4a + 2b + c = 0$
 (3) $ax^2 + x(b - 4a) + 4a + 2b + c = 0$
 (4) $ax^2 + x(b - 4a) + 4a - 2b + c = 0$
- The integral value of a for which the equation $(x + 2010)(x + a) + 1 = 0$ has integral roots is
 (1) 2008 (2) 2010
 (3) 2012 (4) 2014
- If a, b are non-zero real numbers, and α, β the roots of $x^2 + ax + b = 0$, then
 (1) α^2, β^2 are the roots of $x^2 - (2b - a^2)x + a^2 = 0$
 (2) $\frac{1}{\alpha}, \frac{1}{\beta}$ are the roots of $bx^2 + ax + 1 = 0$
 (3) $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are the roots of $bx^2 + (2b - a^2)x + b = 0$
 (4) $-\alpha, -\beta$ are the roots of $x^2 + ax - b = 0$
- If for a positive integer n , the quadratic equation $x(x+1) + (x+1)(x+2) + \dots + (x+n-1)(x+n) = 10n$, has two consecutive integral solutions, then n is equal to:
 (1) 9 (2) 10
 (3) 11 (4) 12
[Challenger]



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (7)
2. (14)
3. (3)
4. (3)
5. (3)

6. (4)
7. (4)
8. (1, 3)
9. (2, 3)
10. (3)



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